

Gonzalo Joaquín Esparza

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EDUCATION

- Massachusetts Institute of Technology** September 2022 – May 2027 (Anticipated Graduation)
- M.Eng., Electrical Engineering and Computer Science – expected May 2027
 - * Thesis: Compact EUV Light Source by Plasmonic Field Enhanced High Harmonic Generation
 - * Advisor: Doctor Keathley, Quantum Nanostructures Nanofabrication Group
 - B.S., Electrical Engineering and Computer Science and B.S., Chemical Engineering – received May 2026
 - Relevant Coursework: Applied Quantum Physics, Optics and Photonics, Physics for Solid-State Applications, Engineering for Impact, Semiconductor Electronic Circuits, Nano Processing Technology, Software Design & Construction, Algorithms, Computation Structures, Engineering, Structures, Machine Learning, Organic Chemistry, Thermodynamics, Fluid Mechanics, Transport Processes
 - GPA: 4.7/5.0

WORK EXPERIENCE

- Quantum Nanostructures and Nanofabrication Group** | *Research Assistant* June 2026 – May 2027
- Developing a process for the consistent fabrication of an optical device with a critical dimension 25 nanometers
 - Developing an optical and electrical setup, in and out of vacuum to characterize device performance
- Analog Devices Inc.** | *Process Engineer Intern* June 2025 – May 2026
- Engineered fabrication processes in MIT.nano cleanroom for consistent device results
 - Simulated implantation with TRIMS to maximize desired characteristics.
 - Performed contractor and supplier outreach to perform critical processing steps and procure raw materials
 - Developed data extraction and visualization program to verify the specifications of commercial off the shelf components
- The Hammond Lab** | *Undergraduate Research Assistant* February 2023 – December 2023
- Synthesized and tested organic polymers as chemical hemostats for severe internal bleeding for studies conducted at Massachusetts General Hospital
 - Characterized and optimized immunotherapeutic nanoparticles used in treatment of ovarian cancer

PROJECTS

- Coherent EUV Source** | *Lithography, PVD, CVD, Diffusion, AFM, SEM* June 2026 – May 2027
- Utilizing electron beam lithography to expose grating features with critical dimension of 25 nm
 - Characterizing grating quality using scanning electron microscopy and atomic force microscopy
 - Engineering device geometry to maximize light generation and efficiency
- Novel High Voltage Transistor** | *Lithography, PVD, Diffusion, SEM, TEM* June 2025 – May 2026
- Optimized metallization parameters to minimize contact resistance
 - Utilized rapid thermal annealing to adjust optimize interface interaction between different metals and substrate
- Park Occupancy Monitor** | *Circuit Design, PCB layout, Power Management* February 2026 – May 2026
- Developed and prototyped infrared break-beam turnstile concept
 - Evaluated power consumption and established power budget
 - Constructed and deployed eight systems at Hoyt Park
- Amplifier Integrated Circuit** | *Circuit Design, Semiconductor Layout* February 2026 – May 2026
- Designed amplifier circuit and laid out a corresponding semiconductor layout
 - Passing DRC and LVS with 29.6dB gain, 176.2 μ W consumption, and 365.5ps propagation delay

SKILLS

Programming Languages: Python (gdsfactory, numpy, matplotlib), MATLAB, TypeScript, C
Fabrication: Optical and Electron Beam Lithography, Physical Deposition, Chemical Deposition, Thermal Diffusion
Characterization: Scanning Electron Microscopy, Atomic Force Microscopy, Transmission Electron Microscopy
Software: KLayout, KiCad, Cadence Virtuoso, Git, Bash, VS Code